

LLMs and Prompt Engineering

Let's begin the discussion by answering a few questions.

LLM and Prompt Engineering Quiz

Which factor is the backbone of GPT-style LLM models, using self-attention to weigh the importance of different words?

A

Fine-Tuning

B

Transformer

C

Compute Power

D

Diverse Sources



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What Is a Large Language Model?

A large language model (LLM) is a foundation model trained on huge volumes of text to predict and generate language. Different LLMs are tuned toward different strengths:

Model	Known For
GPT-4	Best for creative content
Falcon	Best for conversation
Llama 2	Free, resource-light, customization
Gemini	Google Assistant
Copilot	Microsoft Assistant
Claude 3	Large context
BERT	Understanding

Foundation models

LLMs are one application of foundation models — huge models trained once on broad data, then applied to many downstream tasks: answering questions, search, summarization, captioning, and more, all steered through a prompt.



LLM and Prompt Engineering Quiz

A model confidently describes detailed, specific facts about a person who never existed. This is an example of:

A

Generalization

B

Fabrication (a form of hallucination)

C

Fine-tuning

D

Active prompting



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Why Does an LLM Appear Smart?

Transformer

Backbone of GPT models. Self-attention weighs the importance of different words to understand context, and allows parallel processing.

Large-Scale Pretraining

Learns to recognize patterns in text — grammar, facts — and builds an understanding of how language is structured.

Fine-Tuning

Supervised learning on specific datasets using human annotators; reinforcement learning improves performance further.

Compute Power

GPU and TPU processing of terabytes of data and billions of parameters.

Optimization

Optimizes a loss function using gradient descent, adjusting parameters through backpropagation of errors.

Contextual Embedding

Numerical representation of words that captures context, letting the model generate relevant responses to new inputs.

Diverse Sources

Books, articles, websites, and more provide a broad base of linguistic knowledge.



Prompt Engineering

Prompt: A **piece of text** or a **set of instructions** that is given to a Generative AI model to **elicit a response** or **accomplish a task**

Prompts play a **critical role** in determining **the quality and relevance of the output** generated by the Generative AI model

Engineering: Iteratively deriving a specific prompt for the task

Example

Task:

Identify the language of the following sentence

Prompt:

Identify the language of the following sentence:
"Bonjour, comment allez-vous ?"

Output:

French

LLM and Prompt Engineering Quiz

Which of the following statements best describes what a 'prompt' is?

A

A mathematical function used in algorithms

B

A piece of text that guides the response of an AI model

C

A data structure used to store model parameters

D

A type of dataset for training models



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LLM and Prompt Engineering Quiz

Consider a scenario where you want to review customer feedback data for a product. Which prompt is likely to yield the most valuable output?

A

"Analyze the customer feedback data"

B

"Give your suggestions on the customer feedback data"

C

"You're a Product Manager. Analyze the customer feedback and recommend product changes to improve customer satisfaction."

D

"What do you think about the customer feedback from the data provided?"



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Specifying a persona for the model

Specifying a persona or role for the model, such as **“Product Manager”**, **“Data Analyst”**, or **“AI Instructor”** helps align its responses with the required expertise.

The model often produces **generic or unfocused responses** due to their broad training, which **can result in a lack of task-specific alignment**.

Assigning a specific role to the model **ensures more contextually relevant and accurate outputs**, improving alignment with task requirements

Example: "As a financial analyst, evaluate this investment risk by considering market trends, company performance, and external factors. Then, provide your conclusion."



Chain-of-Thought Prompting

Encourages step-by-step reasoning before arriving at a clear, structured output

Excels in tasks requiring **logical deduction, multi-step problem-solving, or eliminating incorrect possibilities**

Can be unnecessary or overkill for straightforward classification or extraction tasks

Most beneficial for reasoning-intensive tasks

Multi-step reasoning tasks

E.g., Diagnosing a disease by considering symptoms and eliminating unlikely conditions

Math and logical problems

E.g., Solving word problems where multiple calculations are required



Chain-of-Thought Prompting: Examples

Directly instruct the model to follow a structured reasoning process

"Solve the problem step by step. First, identify key information, then analyze possible solutions, and finally conclude with the best answer."

Ask the model to first outline the reasoning steps before solving the problem

"Before answering, list the steps you would take to solve this problem. Then, apply those steps to find the answer."

Ask the model to first evaluate common mistakes before solving the problem

"Before solving, identify common errors people make in this type of problem. Then, provide a correct solution."

Ask the model to generate multiple possible answers, assess their validity, and then pick the best one

"List all possible answers, analyze why each might be correct or incorrect, and then choose the best one with justification."



LLM and Prompt Engineering Quiz

What is the main idea behind Chain-of-Thought prompting?

A

Break a complex task into a series of simpler steps, guiding the model through a chain of reasoning

B

Provide the model with several examples of the task before asking it to perform it

C

Explore multiple branching reasoning paths and evaluate each before converging on a solution

D

Have the model proactively ask the user for clarification before proceeding



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Where Chain-of-Thought Prompting is an Overkill?

Simple information extraction tasks

E.g., Extracting names and dates from a document

Binary classification with clear patterns

E.g., spam vs. not spam in emails

Tasks where immediate recall is sufficient

E.g., Retrieving a factual answer from a knowledge base

When concise responses are required

E.g., Generating short product descriptions

Real-time applications where speed is critical

E.g., Chatbots responding instantly to customer queries



Power Ahead !

